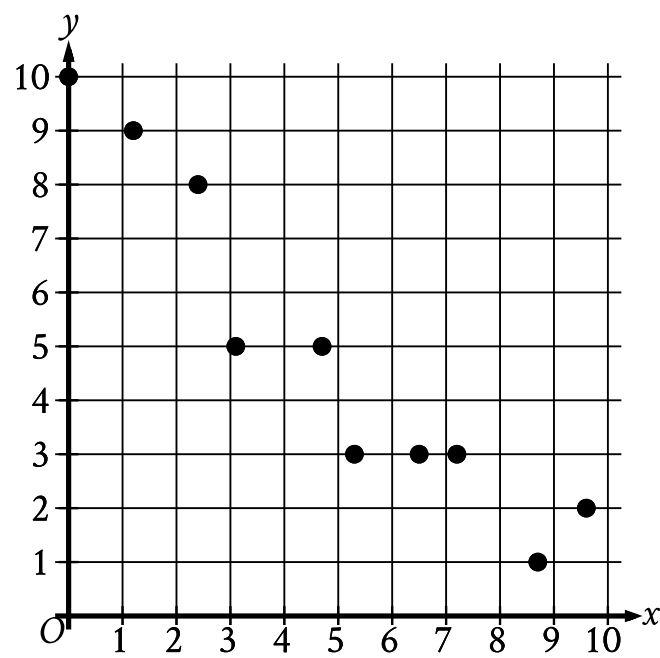


| Assessment | Test | Domain                            | Skill                                      | Difficulty                                          |
|------------|------|-----------------------------------|--------------------------------------------|-----------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Medium <div><div></div><div></div><div></div></div> |

Question

The scatterplot shows the relationship between two variables,  $x$  and  $y$ .



Which of the following equations is the most appropriate linear model for the data shown?

Answer

- A.  $y = 0.9 + 9.4x$
- B.  $y = 0.9 - 9.4x$
- C.  $y = 9.4 + 0.9x$
- D.  $y = 9.4 - 0.9x$

Correct Answer: D

Rationale

Choice D is correct. The data points suggest that as the variable  $x$  increases, the variable  $y$  decreases, which implies that an appropriate linear model for the data has a negative slope. The data points also show that when  $x$  is close to  $0$ ,  $y$  is greater than  $9$ . Therefore, the  $y$ -intercept of the graph of an appropriate linear model has a  $y$ -coordinate greater than  $9$ . The graph of an equation of the form  $y = a + bx$ , where  $a$  and  $b$  are constants, has a  $y$ -intercept with a  $y$ -coordinate of  $a$  and has a slope of  $b$ . Of the given choices, only choice D represents a graph that has a negative slope,  $-0.9$ , and a  $y$ -intercept with a  $y$ -coordinate greater than  $9$ ,  $9.4$ .

Choice A is incorrect. The graph of this equation has a positive slope, not a negative slope, and a  $y$ -intercept with a  $y$ -coordinate less than  $1$ , not greater than  $9$ .

Choice B is incorrect. The graph of this equation has a  $y$ -intercept with a  $y$ -coordinate less than **1**, not greater than **9**.

Choice C is incorrect. The graph of this equation has a positive slope, not a negative slope.